
ASSIGNMENT 21

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Domain: AIML (Artificial Intelligence and Machine Learning)

<https://github.com/omkulawade03/Frontend-Session-Assignments.git>

Q1. Setup and Libraries

Import the required libraries.

Program Code:

#Questions 01

```
app.py > ...
1  # =====
2  # Q1. Setup and Libraries
3  # Import the required libraries.
4  # =====
5
6  import streamlit as st
7  import pandas as pd
8  import joblib
9
```

OUTPUT SCREENSHOT:



```
PS C:\Users\Asus\OneDrive\Desktop\Fronted_Learning\Amazon_Price_Predictor> .& "C:\Users\Asus\AppData\Local\Python\pythoncore-3.14.6\python.exe"
^ "C:\Users\Asus\.vscode\extensions\ms-python.debugpy-2026.6.0-w32-x64\handed\libs\debugpy\launcher" "62893" "..." "C:\Users\Asus\OneDrive\De
sktop\Fronted_Learning\Amazon_Price_Predictor\app.py"
```

Q2. Loading Model and Preprocessing Objects

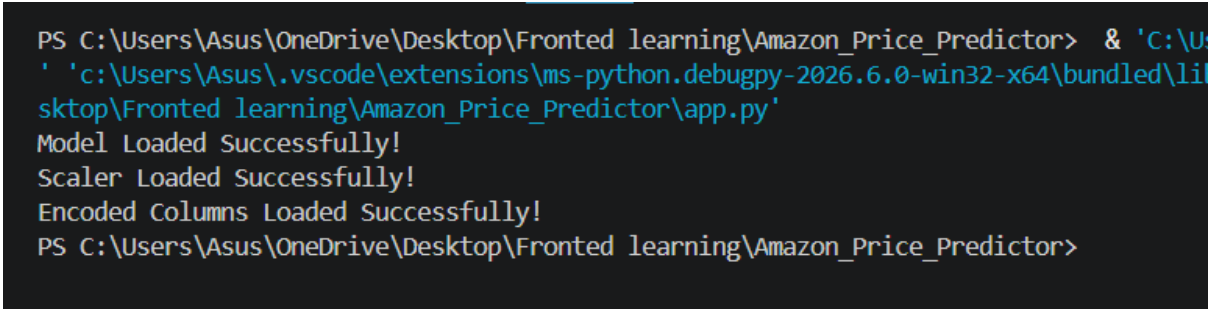
Load the trained model, scaler and encoded columns.

Program Code:

#Questions 02

```
10
11  # =====
12  # Q2. Loading Model and Preprocessing Objects
13  # Load the trained model, scaler and encoded columns.
14  # =====
15
16  # Load Trained Machine Learning Model
17  model = joblib.load("amazon_model.pkl")
18  print("Model Loaded Successfully!")
19
20  # Load Standard Scaler
21  scaler = joblib.load("amazon_scaler.pkl")
22  print("Scaler Loaded Successfully!")
23
24  # Load Encoded Column Names
25  columns = joblib.load("amazon_columns.pkl")
26  print("Encoded Columns Loaded Successfully!")
27
```

OUTPUT SCREENSHOT:



```
PS C:\Users\Asus\OneDrive\Desktop\Fronted learning\Amazon_Price_Predictor> & 'C:\U
' 'c:\Users\Asus\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\li
sktop\Fronted learning\Amazon_Price_Predictor\app.py'
Model Loaded Successfully!
Scaler Loaded Successfully!
Encoded Columns Loaded Successfully!
PS C:\Users\Asus\OneDrive\Desktop\Fronted learning\Amazon_Price_Predictor>
```

Q3. Page Configuration

Program Code:

#Questions 03

```
29
30  # =====
31  # Q3. Page Configuration
32  # =====
33
34  st.set_page_config(
35      page_title="Amazon Product Price Predictor",
36      page_icon="🛒",
37      layout="centered"
38  )
39
```

OUTPUT SCREENSHOT:

```
PS C:\Users\Asus\OneDrive\Desktop\Fronted learning\Amazon_Price_Predictor> python -m streamlit run app.py
2026-07-18 14:46:11.201 Uvicorn server started on :::8501

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://192.168.1.34:8501

Help agents write better Streamlit apps?
Install the official Streamlit skills by running streamlit skills in your terminal.

Model Loaded Successfully!
Scaler Loaded Successfully!
Encoded Columns Loaded Successfully!
█
```

Q4. Title and Description

Program Code:

#Questions 04

```
40
41  # =====
42  # Q4. Title and Description
43  # =====
44
45  st.title("Amazon Product Price Predictor")
46
47  st.write("Enter the product details below to predict the discounted price.")
48
49  st.write("---")
50
```

OUTPUT SCREENSHOT:



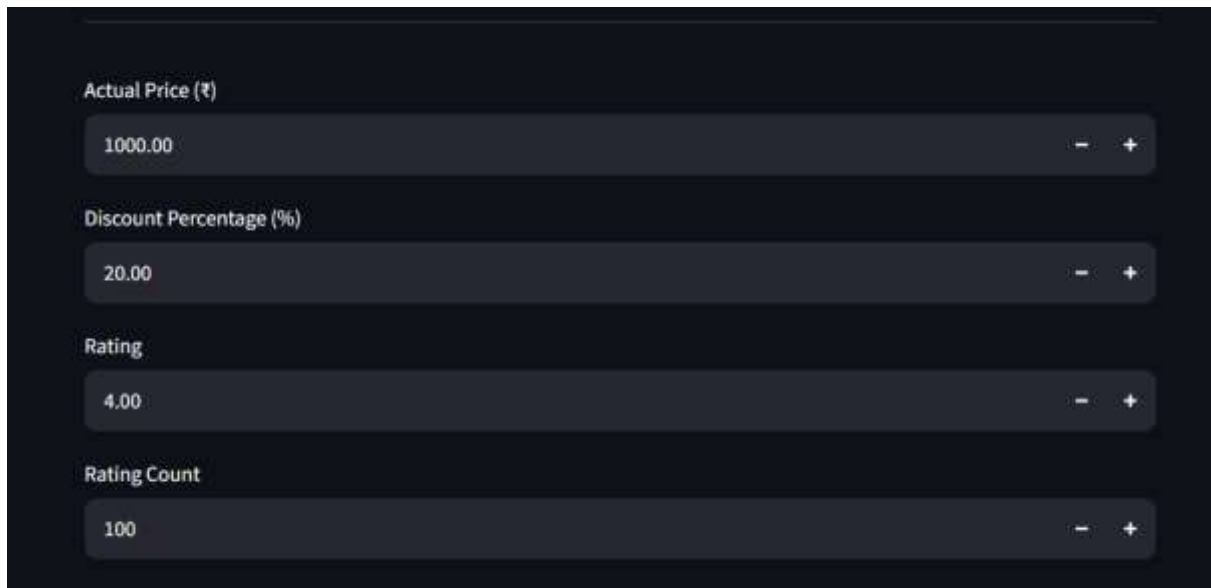
Q5. Numerical Input Fields

Program Code:

#Questions 05

```
53  # =====
54  # Q5. Numerical Input Fields
55  # =====
56
57  actual_price = st.number_input(
58      "Actual Price (₹)",
59      min_value=1.0,
60      value=1000.0
61  )
62
63  discount_percentage = st.number_input(
64      "Discount Percentage (%)",
65      min_value=0.0,
66      max_value=100.0,
67      value=20.0
68  )
69
70  rating = st.number_input(
71      "Rating",
72      min_value=0.0,
73      max_value=5.0,
74      value=4.0
75  )
76
77  rating_count = st.number_input(
78      "Rating Count",
79      min_value=0,
80      value=100
81  )
82
```

OUTPUT SCREENSHOT:



A screenshot of a dark-themed user interface showing four input fields. Each field consists of a label, a text input area with a value, and minus/plus buttons for adjustment.

Field Label	Value	Buttons
Actual Price (₹)	1000.00	- +
Discount Percentage (%)	20.00	- +
Rating	4.00	- +
Rating Count	100	- +

Q6. Categorical Input using Dropdowns

Program Code:

#Questions 06

```
83  # =====
84  # Q6. Categorical Input using Dropdowns
85  # =====
86
87  category = st.selectbox(
88      "Product Category",
89      [
90          "Electronics",
91          "Home & Kitchen",
92          "Computers & Accessories",
93          "Musical Instruments",
94          "Office Products",
95          "Health & Personal Care",
96          "Home Improvement",
97          "Car & Motorbike"
98      ]
99  )
100
101  brand = st.selectbox(
102      "Brand",
103      [
104          "Amazon Basics",
105          "boAt",
106          "Samsung",
107          "Apple",
108          "OnePlus",
109          "HP",
110          "Dell",
111          "Sony",
112          "JBL",
113          "Logitech",
114          "Realme",
115          "Mi"
116      ]
117  )
118
```

```
118  
119     product_type = st.selectbox(  
120         "Product Type",  
121         [  
122             "Mobile Phone",  
123             "Laptop",  
124             "Headphones",  
125             "Smart Watch",  
126             "Keyboard",  
127             "Mouse",  
128             "Power Bank",  
129             "USB Cable",  
130             "Bluetooth Speaker",  
131             "Monitor",  
132             "Printer",  
133             "Charger"  
134         ]  
135     )
```

OUTPUT SCREENSHOT:

The image displays three sequential screenshots of a product selection interface, each showing a dropdown menu with a list of options. The interface has a dark theme.

Product Category

- Electronics
- Home & Kitchen
- Computers & Accessories
- Musical Instruments
- Office Products
- Health & Personal Care
- Home Improvement

Brand

- Amazon Basics
- boAt
- Samsung
- Apple
- OnePlus
- HP
- Dell

Product Type

- Mobile Phone
- Laptop
- Headphones
- Smart Watch
- Keyboard
- Mouse
- Power Bank

Q7. Predict Button

Program Code:

#Questions 07

```
136  
137     # =====  
138     # Q7. Predict Button  
139     # =====  
140  
141     predict = st.button("Predict Price")  
142
```

OUTPUT SCREENSHOT:

A screenshot of a web application interface for predicting product prices. The interface is dark-themed and contains several input fields and a button. The inputs are: a rating field set to 4.00, a rating count field set to 100, a product category dropdown set to Electronics, a brand dropdown set to Amazon Basics, and a product type dropdown set to Mobile Phone. A Predict Price button is located at the bottom of the input section.

Field	Value
Rating	4.00
Rating Count	100
Product Category	Electronics
Brand	Amazon Basics
Product Type	Mobile Phone

Predict Price

Q8. Creating Input DataFrame

Program Code:

#Questions 08

```
144 # =====
145 # Q8. Creating Input DataFrame
146 # =====
147
148 if predict:
149
150     st.subheader("Product Details")
151
152     display_df = pd.DataFrame({
153         "Category": [category],
154         "Brand": [brand],
155         "Product Type": [product_type],
156         "Actual Price": [actual_price],
157         "Discount %": [discount_percentage],
158         "Rating": [rating],
159         "Rating Count": [rating_count]
160     })
161
162     st.dataframe(display_df)
163
164     # DataFrame used for prediction
165     input_df = pd.DataFrame({
166         "actual_price": [actual_price],
167         "discount_percentage": [discount_percentage],
168         "rating": [rating],
169         "rating_count": [rating_count]
170     })
171
```

OUTPUT SCREENSHOT:

Actual Price (₹)

1000.00

-

+

Discount Percentage (%)

20.00

-

+

Rating

4.00

-

+

Rating Count

100

-

+

Product Category

Electronics

▼

Brand

Amazon Basics

▼

Product Type

Mobile Phone

▼

Predict Price

Product Details

	Category	Brand	Product Type	Actual Price	Discount %	Rating	Rating Count
0	Electronics	Amazon Basics	Mobile Phone	1000	20	4	100

Q9. Feature Scaling and Prediction

Program Code:

#Questions 09

```
172 # =====
173 # Q9. Feature Scaling and Prediction
174 # =====
175
176     input_scaled = scaler.transform(input_df)
177
178     prediction = model.predict(input_scaled)
179
180     st.success(
181         f"Predicted Discounted Price: ₹{prediction[0]:,.2f}"
182     )
```

OUTPUT SCREENSHOT:

Predict Price

Product Details

	Category	Brand	Product Type	Actual Price	Discount %	Rating	Rating Count
0	Electronics	Amazon Basics	Mobile Phone	1000	20	4	100

Predicted Discounted Price: ₹1,679.98

Amazon Product Price Predictor

Enter the product details below to predict the discounted price.

Actual Price (₹)

1500000.00

-

+

Discount Percentage (%)

30.00

-

+

Rating

4.00

-

+

Rating Count

300

-

+

Product Category

Electronics

▼

Brand

Apple

▼

Product Type

Laptop

▼

Predict Price

Product Details

	Category	Brand	Product Type	Actual Price	Discount %	Rating	Rating Count
0	Electronics	Apple	Laptop	1500000	30	4	300

Predicted Discounted Price: ₹910,092.39